

THERE IS NO LIGHT AT THE END OF THE TUNNEL OF LIFE

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SURVIVORSHIP BIAS IN IDL DATA

Numerous studies tried to address the existence of an upper bound to human longevity by analyzing data about how much people have lived in the past.^[1] With currently available and reliable datasets of mortality at high ages, this approach inevitably leads to the estimation of a constant hazard and an exponentially distributed excess lifetime above ages around 105, implying an unlimited lifetime. In fact, by performing a Kaplan-Meier estimate of the hazard function from the most recent version of the IDL dataset, we found a well-expected constant hazard rate, with an annual probability of death consistently around 0.5. We explain this with a **survivorship bias**: only people with an extraordinary resistance to ageing reach ages above 105 years, generating a highly selected sample that due to genetical and environmental factors is very resistant to ageing. Therefore, for this sample the risk of mortality increases at a slower rate year after year.

LTL: AN ALTERNATIVE APPROACH

However, we propose in this article an alternative approach: instead of trying to infer for how many consecutive years it is possible to stay alive, we reframe the problem into trying to understand the maximum level of ageing a person can experience before dying.

For this purpose, we analyzed a dataset about **Leukocytes Telomeres Length (LTL)**, a well-established biomarker of ageing associated with deadly diseases in old people, from a 2021 study conducted by the University of Calabria.^[2]

The dataset, of over 500 Italians, reported their LTL (measured in T/S) and their excess lifetime after the measurement. The mean T/S was found to be 0.79 for deceased people, and 1.15 for those who survived ($p < 0.0001$ from the z-test). The Kaplan-Meier estimates (**Figure 1**) of the survival function by quartile of T/S further validate the association between LTL and mortality.

SURVIVORSHIP BIAS IN LTL DATA

To confirm the validity of this approach, we found the same, previously mentioned, survivorship bias in the LTL data as well. Evidence of the survivorship bias lay in the fact that the average LTL of people starts increasing around the age of 90 (**Figure 2**), despite longitudinal data from the study showing that the individual LTL, in the long term, monotonically decreases with age. In fact, in the follow-up sample (36 subjects, mean age of 85.3 ± 3.9), people who died experienced on average 4 times more shortening

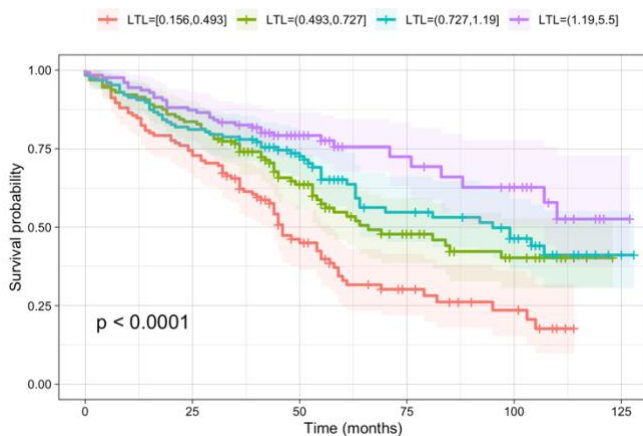


Figure 1. Kaplan-Meier survival curve by T/S quartiles. The curve shows how the KM survival changes depending on the T/S.

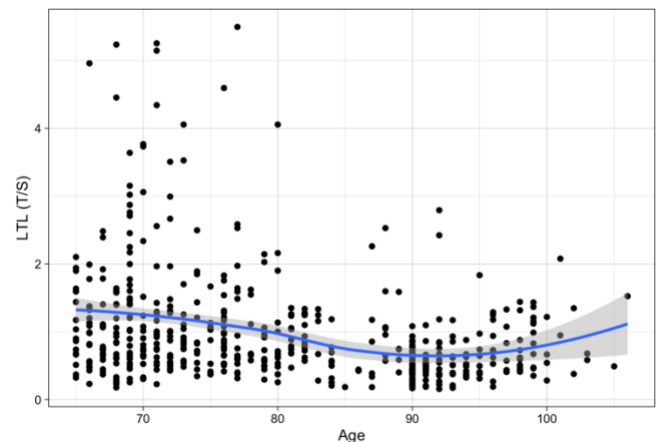


Figure 2. LOESS regression (span=1) on T/S and Age scatterplot. This graph shows the survivorship bias in our LTL dataset.

compared to people who survived (0.23 versus 0.06). These findings suggest the existence of an **upper bound** to the LTL of a person at high ages and that this upper bound, in every subject taken individually, in the long term, monotonically decreases with age.^[3]

A LOWER BOUND TO THE LTL

A meta-analysis by L. Rode et al.^[4] analyzed data about the LTL of more than 64,000 Danish people. Their results, shown in **Figure 3**, provide strong evidence of the existence of a constant level of LTL below which life is no longer possible.

Moreover, such **lower bound** appears to be constant at every age, around 2500 base pairs: this result is consistent with LTL being a predictor of mortality independently of age. We explain the upper outliers in the scatterplot by the association between extremely long telomeres and death from cancer reported in the meta-analysis. This is the reason why the upper outliers tend to disappear at very high ages.

THE TUNNEL OF LIFE

These findings combined with ours suggest the existence of a **“tunnel of life”**: a bounded range that sets the values of LTL between which life is possible at extreme ages.

This tunnel of life has a constant lower bound (as shown by the meta-analysis) and a monotonically decreasing upper bound, as the LTL of the individual itself is monotonically decreasing.

Further statistical evidence suggesting the existence of a tunnel of life is given by an analysis of the distribution of LTL for age quartiles.

The density plot shown in **Figure 4** illustrates that, as age increases, the distribution becomes more concentrated, with a standard deviation of 1.05 in the first quartile and 0.43 in the fourth, suggesting a shrinking of the tunnel of life as it approaches the lower bound. Moreover, the mean LTL decreases from 1.31 in the first quartile to 0.69 in the fourth, reflecting a decreasing upper bound.

Since the tunnel of life has a monotonically decreasing upper bound and a constant lower bound, such a tunnel will eventually close, making it impossible to live past a certain age defined by the intersection of the two bounds.

And this is why there's no light at the end of the tunnel of life. Thus, **life is limited**.

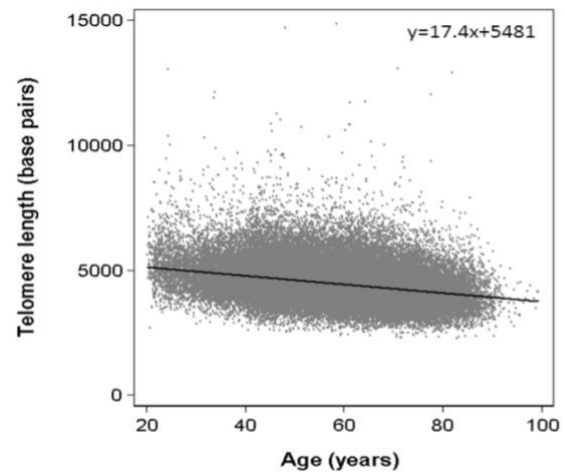


Figure 3. Scatterplot of LTL (measured in base pairs) as a function of age. The scatterplot delimitates the lower bound for LTL, the same independently of age.^[4]

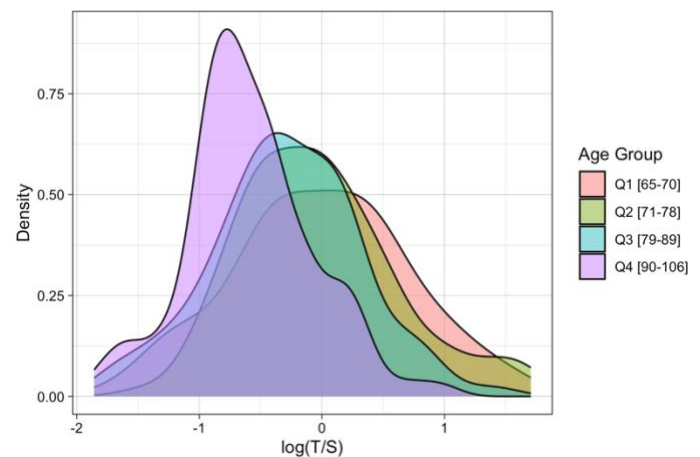


Figure 4. Density plot of LTL (log scale, measured in T/S) by quartile of Age. It shows how LTL range shrinks as people age.

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